



Research article

Interface-engineered novel PPy-TiO₂-SnO₂ ternary nanocomposite for high-sensitivity low-concentration CO₂ gas sensingAditya V. Tiwari^{a,*}, Ashitosh P. Deshmukh^b, Shailesh P. Tiwari^c, Payal S. Dudhe^d, Shikhil S. Wanjar^a, Deepika S. Yawale^b, Shrikrishna P. Yawale^b^a Department of Physics, Shri Mathuradas Mohota College of Science, Nagpur, Maharashtra 440024, India^b Materials Research Laboratory, Department of Physics, Government Vidarbha Institute of Science and Humanities, Amravati, Maharashtra 444604, India^c Department of Physics, N. M. Institute of Science, Bhavan's College, Andheri (W), Mumbai, Maharashtra 400058, India^d Department of Chemistry, Shri Mathuradas Mohota College of Science, Nagpur, Maharashtra 440024, India

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ABSTRACT

Conducting polymer polypyrrole (PPy) and its metal oxide composites, namely PPy-TiO₂ and PPy-SnO₂, were synthesized via in-situ chemical oxidative polymerization with 10% (w/v) metal oxide loading. To advance beyond conventional bi-composites, a new synergistic tri-composite sensor PPy-TiO₂-SnO₂ was developed through ex-situ mixing of the two binary composites. UV-Visible spectroscopy was employed to determine the optical band gaps, while SEM revealed granular, densely connected morphologies, further supported by elemental confirmation from EDX. FTIR analysis verified the interaction between PPy and the incorporated metal oxides, and TGA established the thermal sustainability of the composites for room-temperature sensing applications. Impedimetric analysis showed a linear 45° complex impedance behaviour and confirmed that AC conductivity decreases with increasing bandgap. All composite films were fabricated on glass substrates and evaluated for CO₂ sensing using a laboratory-developed sensing unit. Linear, dynamic, and static responses were recorded across varying CO₂ concentrations. The bi-composites displayed sensitivities of 3.43 (PPy-SnO₂) and 14.15 (PPy-TiO₂), whereas the newly synthesized PPy-TiO₂-SnO₂ exhibited an intermediate yet enhanced sensitivity of 8.09. The tri-composite also demonstrated rapid gas-sensing performance, with a short recovery time of 76 s and response time of 184 s. Energy-level alignment and Fermi-level pinning effects correlated well with the bandgap values, supporting the proposed sensing mechanism involving interfacial depletion and accumulation layers. Overall, the PPy-TiO₂-SnO₂ tri-composite emerges as a robust, sustainable, and efficient room-temperature CO₂ sensing material. The sensor also exhibits good selectivity towards CO₂ over interfering gases such as NH₃, ethanol, and CO. Furthermore, it demonstrates excellent long-term stability with a sensitivity drift of 6.10% over 35 days and minimal baseline variation. The sensor performance remains largely unaffected under varying humidity conditions (30–70% RH), indicating its suitability for practical environmental applications.

1. Introduction

Environmental pollution has emerged as a critical issue due to increased industrial activity, urbanization, vehicular emissions, fossil fuel combustion, and wildfires. [1–3] The continuous rise in atmospheric carbon dioxide (CO₂) concentrations has become a central concern for both environmental sustainability and public health. Increased indoor CO₂ concentrations are increasingly linked to inadequate ventilation, reduced air quality, and adverse impacts on human

cognition. [4] Likewise, outdoor CO₂ emission significantly contributes to climate change, rendering efficient and low-power CO₂ monitoring essential for environmental management, industrial safety, and smart buildings. [5] Traditional gas sensors based on metal oxide semiconductors (MOS) often require high operating temperatures to reach adequate sensitivity and response times, which in turn demands significant energy input and restricts their broader deployment in portable or low-power systems. [6,7]

Conducting polymers (CPs) have emerged as promising materials for

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room-temperature chemiresistive sensing due to their tunable electrical conductivity, ease of synthesis, and strong interaction with analyte molecules. [8] Within the broad category of conductive polymers, polypyrrole (PPy) offers a unique combination of chemical and physical properties, it can be synthesized via low-temperature oxidative polymerization, exhibits good environmental stability, and supports efficient charge transport through a conjugated π -electron backbone which make PPy well suited for room-temperature chemiresistive sensing applications. [9,10] When polypyrrole (PPy) is combined with inorganic phases such as metal oxides, the resulting hybrid materials exploit strong synergistic effects: the polymer provides a soft, conductive matrix that improves charge transport and mechanical integrity while the oxide supplies high surface area and active adsorption sites, together enhancing chemiresistive sensing at or near room temperature. [11]

Titanium dioxide (TiO₂) and tin dioxide (SnO₂) are widely used metal oxides in chemiresistive gas sensors because their surface chemistry is well characterized, they exhibit multiple accessible oxidation states, and they readily form oxygen vacancies that act as active sites to enhance adsorption and charge-transfer processes. [12] While TiO₂ is valued for its chemical stability and favourable band structure [13], SnO₂ is highly active for gas adsorption and charge exchange. [14] However, pure TiO₂ and SnO₂ sensors often require elevated temperatures to operate efficiently. By forming composites with PPy, one can leverage p-n heterojunction formation between the polymer (p-type) and oxide (n-type), creating depletion and accumulation layers at the interfaces. [15] These interfaces modulate charge transport during gas exposure, enhancing sensitivity and enabling operation at much lower temperatures.

Recent literature reinforces the value of PPy-metal oxide composites for sensing applications. Hybrid PPy-TiO₂ nanostructures have shown enhanced optical and electrical properties for catalyst and sensor applications. In parallel, PPy-SnO₂ systems have been demonstrated to improve response and stability compared to pristine PPy. In a representative study, Farea *et al.* fabricated a room-temperature CO gas sensor based on chemically synthesized PPy-TiO₂ nanocomposites and showed that incorporation of TiO₂ into the PPy matrix led to almost a four-fold enhancement in sensitivity compared with pristine PPy, together with faster response and recovery times and good long-term stability; these improvements were attributed to the formation of an efficient p-n heterojunction and enhanced interfacial charge transfer in the composite. [16] Earlier gas-sensing research by Bulakhe *et al.* on PPy-TiO₂ heterojunction films for LPG detection at ambient temperature demonstrated that these hybrids exhibit superior response and faster kinetics compared to the individual components, attributed to the intrinsic electric field at the PPy-TiO₂ interface. [17] Huiling Tai *et al.* discussed PPy-TiO₂ ammonia sensors again describing that Compared with pure PPy thin-film sensors, the PPy-TiO₂ film gas sensor has a shorter response and recovery time. [18] All these studies established PPy-TiO₂ as a robust platform for low-temperature gas sensing.

Hybrid PPy-SnO₂ nanostructures have recently been highlighted as highly effective materials for gas-sensing applications because the incorporation of SnO₂ nanoparticles into a conductive PPy matrix enables multiple synergistic mechanisms. As shown by Nascimento *et al.* SnO₂ exhibits adjustable oxygen-vacancy chemistry and high surface reactivity, which, when interfaced with p-type PPy, leads to significant modulation of electronic conductivity upon gas exposure. [19] Reviews by Husain *et al.* further emphasize that PPy provides a soft, highly conductive framework that ensures continuous charge-transport pathways, while SnO₂ contributes abundant adsorption sites and rapid electron-exchange behaviour enabling enhanced sensitivity at near room temperature. [20] Moreover, the trend toward multicomponent composites has grown that integrate polypyrrole (PPy) with two inorganic metal-oxide phases which have begun to attract attention because they combine complementary adsorption chemistries and electronic properties into a single sensing layer. The ternary systems for example PPy-SnO₂-graphene nanoribbons prepared by Hsieh *et al.* (2021) have demonstrated that introducing carbonaceous phases further improves

sensor response and stability due to enhanced charge mobility and increased surface area. [21] These collective findings, reinforced by broader analyses of SnO₂ hybrid nanostructures consistently indicate that PPy-SnO₂ composites leverage interfacial heterojunction effects, oxygen-vacancy driven adsorption, and morphological tuning to deliver high-performance low-temperature gas sensing. In a ternary films of PPy-TiO₂-ZnO synthesized by Chowdhury *et al.* demonstrated improved electrochemical sensing of nitrite and enhanced electrochemical performance compared with PPy or binary mixes, which the authors attributed to stronger interfacial coupling and faster charge transfer across multiple heterojunctions in material. [22] Similarly Muthusamy *et al.* developed PPy-Prussian blue-TiO₂ hybrids which have been utilized as fiber-optic and thin-film sensors for volatile organics with its ternary structure enhancing the available adsorption and interaction sites, thereby improving signal transduction. [23] Despite this progress, most studies remain limited to binary composites (i.e., PPy-TiO₂ or PPy-SnO₂). There is relatively little exploration of tri-phase systems, where PPy is simultaneously integrated with two complementary metal oxides such as TiO₂ and SnO₂. Such a tri-composite could potentially exploit the individual strengths of both oxides, TiO₂ for stable surface states with band-edge properties, SnO₂ for high adsorption and catalytic reactivity while harnessing enhanced interfacial charge redistribution via multiple heterojunctions.

Recent studies have further highlighted the critical role of interface engineering and hierarchical nanostructuring in enhancing gas sensing performance. A report by Zhimin Yang *et al.* demonstrated that engineered heterojunction isolated Cu-N5 sites engineered polypyrrole-reduced graphene oxide hybrids significantly improve sensing response through enhanced charge separation, increased density of active adsorption sites, and improved carrier transport across interfaces. [24] The study emphasized that modulation of interfacial electronic properties directly governs sensitivity and response kinetics, particularly at room temperature.

Similarly, a recent study reported in Chemical Engineering Journal demonstrated the design of hydrogen bonding-mediated α -Fe₂O₃-decorated reduced graphene oxide (rGO) hybrids for dual-mode humidity sensing. The study highlighted that controlled interfacial interactions and surface functionalization significantly enhance adsorption behavior, enabling rapid response, low hysteresis, and stable signal output. [25] Such advanced material engineering approaches underline the importance of synergistic interactions and interface-driven mechanisms in improving sensing performance. These findings strongly support the strategy adopted in the present work, where a PPy-TiO₂-SnO₂ tri-composite is designed to exploit multiple heterointerfaces and synergistic effects for improved CO₂ sensing performance at room temperature.

The earlier works by S. P. Yawale and colleagues laid important groundwork in this domain. For instance, Waghuley *et al.* demonstrated a PPy-based CO₂ sensor using chemically synthesized PPy films. [26] Their study showed that PPy's resistance changes in response to CO₂ exposure at room temperature, setting a precedent for polymer-based CO₂ sensing. In a follow-up, S. P. Yawale's group also reported a SnO₂/PPy screen-printed multilayer CO₂ sensor, further highlighting how combining PPy with SnO₂ enhances sensing performance. [27]

Building on this foundation, in the present work, we design, synthesize, and characterize a novel PPy-SnO₂-TiO₂ tri-composite and evaluate its performance as a room-temperature CO₂ sensor. We have combined in-situ chemical oxidative polymerization (to produce PPy-oxide binary composites) with ex-situ mechanical mixing to form the tri-composit. The oxide loading in the binary composites is maintained at 10% w/v relative to pyrrole monomer. Thin films of the materials were deposited on glass substrates to fabricate resistive sensors. To understand their optoelectronic behavior, we use UV-visible spectroscopy and Tauc plot analysis to determine optical band gaps, which in turn inform our understanding of interfacial energy levels and charge transport mechanisms.

Our sensing experiments employ a voltage-drop method to capture linear, static, and dynamic responses of the sensors under controlled CO₂ concentrations (in ppm) at room temperature. The tri-composite sensor exhibits a moderate yet significantly improved sensitivity (linear response coefficient at 8.09) compared to PPy-SnO₂ (3.43) and PPy-TiO₂ (14.15), while achieving fast response time 184 s and recovery time of 76 s. These improvements are rationalized through the formation of multiple heterointerfaces (PPy-SnO₂ and PPy-TiO₂), Fermi-level pinning, and local depletion/accumulation layers, leading to more efficient charge transfer and dynamic modulation during CO₂ adsorption and desorption. By linking structural, optical, and electrical data, this work offers a comprehensive mechanistic picture of how a tri-phase PPy-oxide composite can outperform its binary counterparts in CO₂ sensing. It also suggests a promising materials design strategy for next-generation, low-power, room-temperature gas sensors based on conducting polymers and multi-oxide architectures.

2. Experimental

2.1. Materials and Chemicals

Extra pure Pyrrole monomer procured from Himedia, Methanol and Stannic dioxide (SnO₂ 99.9% pure) received from Loba Chemicals, Titanium dioxide (TiO₂ 99.9% pure) obtained from Ultrananotech, anhydrous Ferric Chloride (FeCl₃) A.R. obtained from Himedia used as oxidizing agent for the polymerization of Pyrrole monomer, Whatmann grade one filter paper were used for filtering the product, Distilled water and methanol were used for washing the product. Screen printing hand-machine was used for sensor coating on clinically cleaned glass slides.

2.2. Synthesis of Pure Polypyrrole

For the synthesis of Polypyrrole methanol is taken as a solvent. The ratio of pyrrole monomer to the oxidizing reagent FeCl₃ is taken in ratio of 1:2.33 in such a way to obtain maximum yield of the polypyrrole out of polymerization reaction. [28] Initially 9.01 g of anhydrous FeCl₃ is added into the 40 ml of methanol under constant stirring in a beaker for 10 min. As soon as the FeCl₃ is added into the methanol produces heated fumes and solution becomes warm, showing exothermic reaction. This freshly prepared solution is further ultrasonicated for 30 min in a sonication bath for homogeneous solution of FeCl₃. Using syringe 4 ml Pyrrole monomer is added drop wise into the prepared oxidizing solution slowly for 30 min under constant stirring. Reaction is kept under stirring for 2 hrs. As the methanol evaporates from the reaction mixture the local effective oxidizing strength and monomer/oxidant ratio increases, accelerating the polymerization and producing a dense, dark (greenish-black) polypyrrole paste in the beaker. [29] For approaching the complete polymerization of pyrrole monomer the reaction is kept unagitated for 2 hrs in a reaction beaker. The obtained product was filtered and washed several times with distilled water and methanol to eliminate any residual unreacted pyrrole and the ferric and ferrous chloride by-products generated during the polymerization process. Product was finally dried in an oven (Gallenkamp British made) at 60–70 °C for 12 hrs. Then, the resulted conducting polymer product was taken in a plastic pouch and pressed slowly to obtain fine Greenish-black polypyrrole powder.

2.3. Synthesis of Ppy-TiO₂, Ppy-SnO₂ and Ppy-TiO₂-SnO₂ nanocomposites

We obtained two separate composites by doping Metal oxide TiO₂ and SnO₂ by 10% w/v of a Pyrrole monomer which were initially added with FeCl₃ methanol solution. These PPy-TiO₂ and PPy-SnO₂ nanocomposites were synthesized by dispersing TiO₂ and SnO₂ powders respectively into an FeCl₃ solution at the time of polymerization reaction, followed by 1 h of sonication to ensure uniform dispersion. Further addition of Pyrrole

monomer into the sonicated metal oxide-FeCl₃ solution leads to the polymerization as per the standard procedure to obtain respective nanocomposites. These two composites were then mixed ex-situ using a mortar and pestle under suitable pressure to form a fine greenish-black tricomposite, PPy-TiO₂-SnO₂. This advanced sensing material exhibits modified energy band gap properties due to the effective integration of the two metal oxides within the polypyrrole matrix.

2.4. Development of sensors and their calibration

PPy-TiO₂, PPy-SnO₂ and PPy-TiO₂-SnO₂ sensors were prepared by using screen-printing technique, followed by an adequate heat treatment. To obtain the printing inks of the nanocomposites it is essential to prepare binder which was obtained by thoroughly mixing 8 wt% of butyl carbitol with 92 wt% of ethyl cellulose in agate mortar. This binder was then used to prepare a paste for screen printing by mixing it with 60 wt% of nanocomposite powders and grinding the mixture thoroughly in an agate mortar to ensure uniformity. The resulting pastes were screen-printed onto chemically cleaned glass substrates measuring 75 mm × 25 mm. These printed films were allowed to dry at room temperature (303 K) for more than 24 h. After drying, the films were heated at temperatures between 373 K and 403 K for 3 h in an oven. This heating process helped removing carbitol through decomposition and ensured proper adhesion of the film to the glass substrate. As a result, no solvent diffusion occurred in the printed films.

For surface resistance measurements of the films, silver conductive paste was applied on opposite sides of the films to obtain electrodes for the electrical contacts. The electrodes were formed by applying silver conductive paste along the opposite edges of the glass substrate (25 mm width), resulting in a parallel electrode configuration. The electrode contact width was maintained at approximately 5–6 mm with a uniform coating to ensure stable electrical contact. Although the electrode geometry was not systematically optimized in the present study, this configuration provided consistent and reproducible sensing response. These electrodes were dried by heating the films at 70 °C for 30 min. The electrical resistance of the films were measured using the voltage drop technique which was used for the CO₂ gas sensing purpose in our study. [30] The voltage drop across the resistance connected in series with the sensor sample was calculated. The change in the surface resistance of the sensors in an ambient CO₂ gas environment produced dynamic voltage across the reference resistor which was calculated using micro voltmeter. The circuit used for the gas sensing purpose is adopted according to the voltage drop method shown in Fig. 1.

To test the sensor performance, a gas chamber with internal dimensions of 30 cm × 30 cm × 30 cm was used in which and a gas flow meter (Flowtron, India, 1–10 litre/min) was used. The flow rate was set to 1 litre/min. Sensor readings were taken in a CO₂ atmosphere at various gas concentrations at room temperature. Each experiment was repeated 4–5 times to ensure the reproducibility of the sensor response. Sensor stability in CO₂ gas environment was also tested for 1–2 h by keeping the sensor in a gas chamber. No change in surface resistance was observed during this period, confirming sensor stability. The thickness of the sensor films were measured using a Digimatic Micrometer Screw gauge (Japan, Series-293) having resolution of ±0.001 mm. The measured thicknesses of the thick films formed on the sensor substrates by screen printing were 19 µm, 21 µm, 22 µm and 22 µm for the pure PPy, PPy-TiO₂, PPy-SnO₂ and PPy-TiO₂-SnO₂ respectively.

3. Results and discussion

3.1. X-ray Diffraction Analysis

X-ray powder diffraction (XRD) patterns for all the prepared samples were recorded at the Physics Instrumentation Laboratory, Government Vidarbha Institute of Science and Humanities, Amravati. A Rigaku Miniflex benchtop X-ray diffractometer equipped with an advanced

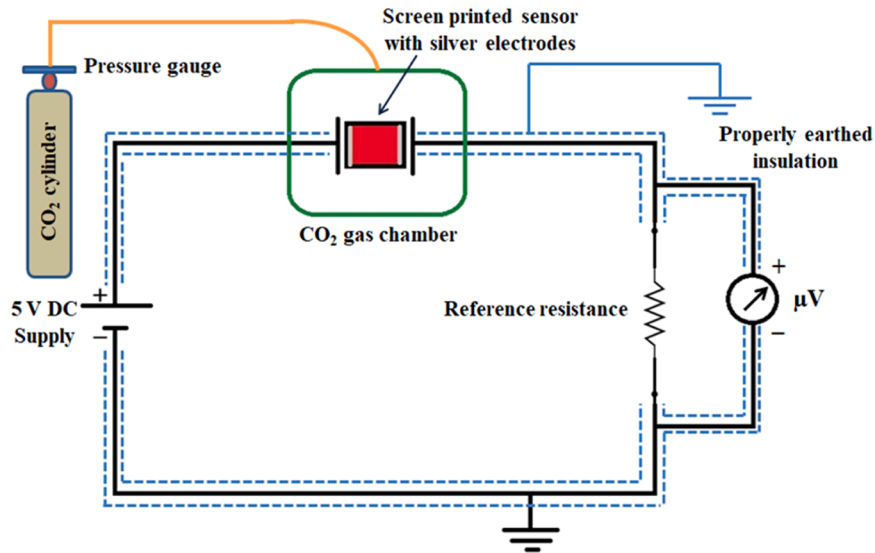


Fig. 1. Schematics of CO₂ gas sensing assembly.

detector system was employed for this purpose. The measurements were carried out using CuK α radiation, with the X-ray generator operated at a tube voltage of 40 kV and a current of 15 mA. Data was collected over a 2θ range of 0° to 80°, with a step size of 13 mm as shown in Fig. 2. A water flow rate of 4.3 L/min was maintained during the measurements to ensure stable operating conditions.

The X-ray diffraction (XRD) pattern of pristine polypyrrole (PPy) exhibits a broad, low-intensity halo in the 2θ region of approximately 20–30°, consistent with an amorphous conducting polymer lacking long-range crystalline order. This diffuse scattering arises from the disordered arrangement of polymer chains and is a characteristic of chemically polymerised PPy films and powders. [31]

Upon incorporation of TiO₂ nanoparticles, the PPy–TiO₂ composite displays a series of sharp diffraction reflections superimposed on the amorphous PPy background, indicating that crystalline TiO₂ is retained within the polymer matrix. The principal reflections appear at approximately 25.3°, 27.3°, 37.9°, 47.9° and near 54.0°, which can be assigned to the (101), (110), (004), (200) and overlapping (105)/(211) planes, respectively; these assignments agree with standard JCPDS references and recent experimental studies of mixed-phase TiO₂ and TiO₂-polymer composites. [32,33] The peaks at 25.3°, 37.9°, and 47.9° are

characteristic of the anatase polymorph, while the peak at 27.3° is characteristic of rutile. [34] The peak near 54.0° occurs in a region where anatase and rutile reflections are closely spaced, [35] and the observed envelope in the composite is therefore reported as an overlapping contribution from anatase (105) and rutile (211) unless peak deconvolution demonstrates a single dominant component. These peak positions and their indexing follow standard TiO₂ reference patterns and recent mixed-phase TiO₂ studies. Anatase and rutile are the two commonly observed TiO₂ polymorphs under ambient synthesis conditions; both adopt tetragonal symmetry but differ subtly in the connectivity of TiO₆ octahedra and unit-cell parameters, which leads to distinct diffraction signatures. [36] Here in PPy–TiO₂ nanocomposite the Anatase phase typically exhibits its most intense reflection at (101) angle of 25.3°, along with (004) and (200) at higher angles, while rutile phase presents a dominant (110) plane near 27.3°. [37] The coexistence of anatase and rutile in TiO₂ can lead to favorable band alignment at the phase interface, which promotes interfacial charge transfer and enhances charge-carrier separation, [38] thereby influencing the electronic and surface energetics of the system which we have further tried to explain through the plausible proposed sensing mechanism.

In the PPy–SnO₂ naocomposite, a series of well-defined diffraction peaks are observed at diffraction angles 2θ equal to 26.7°, 33.9°, 38.1°, 39.1°, 42.7°, 51.9°, 54.9°, 57.9°, 62.1°, 64.9°, 66.1°, 71.3°, and 78.8°. These positions correspond to the characteristic reflections of tetragonal cassiterite SnO₂ having (h,k,l) plane at (110), (101), (200), (211) and higher-order planes [39–41] which are listed in Table 1. This indicates that SnO₂ maintains its intrinsic crystalline structure after composite formation. The presence of multiple high-angle reflections confirms the good crystallinity of the oxide phase in the composite and is consistent with prior reports on PPy–SnO₂ hybrids. Small, systematic shifts in peaks for both PPy–TiO₂ and PPy–SnO₂ reflections toward higher 2θ values and modest peak broadening were observed and can be

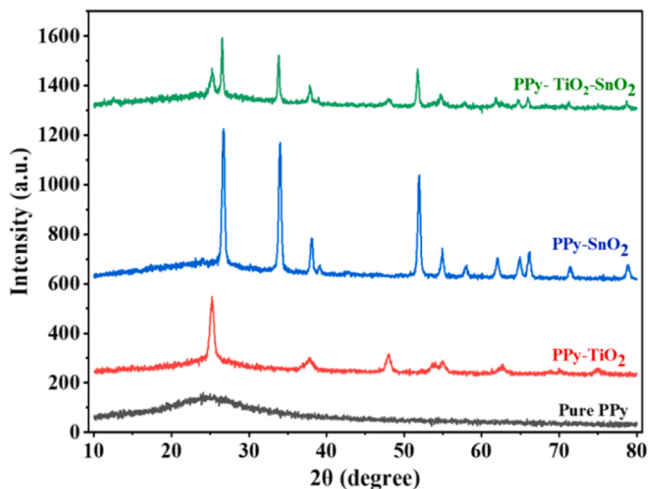


Fig. 2. X-ray diffraction patterns of Pure PPy and its 10% (w/v) doped Composites.

Table 1

Peak position (2θ), FWHM (β), interplanar spacing (d-spacing), crystallite size (D), strain (ϵ), and dislocation density (δ) of synthesized samples.

Sample	2θ (deg)	β (deg)	d- Spacing (Å)	D (nm)	ϵ (10^{-3})	δ (10^{14} 1/ m^2)
TiO ₂	25.27	0.660	3.5215	12.3417	12.8400	65.6523
SnO ₂	26.759	0.398	3.3295	20.5277	7.2976	23.7311
PPy/TiO ₂ / SnO ₂	26.526	0.227	3.3575	35.9741	4.1999	7.7211

rationalized by nanoscale effects at the oxide–polymer interface.

The XRD pattern of the PPy–TiO₂–SnO₂ tricomposite displays the characteristic well-resolved reflections of TiO₂ and SnO₂ superimposed on the broad amorphous halo of polypyrrole which confirms the formation of a hybrid crystalline–amorphous nanocomposite in which oxide nanocrystals are embedded in the PPy matrix. The persistence of sharp oxide reflections indicates that the polymer encapsulation preserved the oxide crystallinity of TiO₂ and SnO₂ within the composite. Such coexistence of crystalline semiconducting oxides (TiO₂, SnO₂) and an amorphous electrically conducting PPy phase promotes intimate oxide–polymer interfaces and is expected to enhance charge transport and surface reactivity.

The most intense peak was used to calculate the crystallite size (*D*) using full width at half maximum (FWHM) and Debye-Scherrer's equation. [42]

$$D = \frac{0.9\lambda}{\beta \cos \theta} \quad (1)$$

where $\lambda = 1.5406 \text{ \AA}$ (CuK α radiation), β is the full width at half maximum (FWHM) in radians, and θ is the Bragg diffraction angle.

The calculated values indicate that the crystallite size (*D*) increases from 12.34 nm for TiO₂ to 20.52 nm for SnO₂, and further to 35.97 nm in the PPy–TiO₂–SnO₂ tricomposite. This progressive increase in crystallite size suggests that during composite formation, the polymer chains of PPy provide a confined growth environment that promotes the coalescence and interconnection of TiO₂ and SnO₂ nanocrystals. The conductive PPy matrix reduces the surface energy of nanoparticles, suppressing uncontrolled nucleation and enabling the formation of larger, more coherent crystallites, which contributes to a more ordered and thermodynamically stable structure.

The following relations were used to compute the microstrain (ϵ) [43] and dislocation density (δ) [44]:

$$\delta = \frac{1}{D^2} \quad (2)$$

$$dx = \frac{8M}{N_A a^3} \quad (3)$$

Simultaneously, the microstrain (ϵ) shows a considerable decrease from 12.84×10^{-3} in TiO₂ to 4.20×10^{-3} in the PPy–TiO₂–SnO₂ tricomposite, suggesting relaxation of internal lattice stress. This can be attributed to the homogeneous dispersion of oxide nanoparticles within the flexible PPy matrix, which effectively accommodates lattice distortions during crystal growth. Likewise, the dislocation density (δ) decreases from 65.65×10^{-14} to $7.72 \times 10^{-14} \text{ m}^{-2}$, indicating enhanced crystallinity and a reduction in structural defects, all calculated values are shown in Table 1. Such improvement in crystalline order promotes better charge carrier mobility and interfacial electron transport within the composite.

The observed increase in crystallite size, along with the decrease in microstrain and dislocation density, confirms that the incorporation of TiO₂ and SnO₂ into the PPy matrix leads to the formation of a highly ordered and defect-minimized hybrid structure. Such structural refinement is beneficial for applications that require efficient charge transport, such as photocatalysis, gas sensing, and energy storage devices.

3.2. FT-IR Analysis

Fourier Transform Infrared (FT-IR) spectra were recorded using a Shimadzu IRAffinity-1S spectrometer, which offers a maximum resolution of 0.5 cm^{-1} . The instrument features a dynamic alignment system with a smoothly moving mirror mechanism, enabling stable and high-quality spectral acquisition with minimal warm-up time. The system samples data at a rate of over 5000 times per second, ensuring optimal performance during measurements. The spectra for the powdered samples were collected in the wavenumber range of $400\text{--}4000 \text{ cm}^{-1}$.

FT-IR spectroscopy confirms the interaction between polypyrrole (PPy) and the incorporated metal oxides, TiO₂ and SnO₂. As shown in Fig. 3, the characteristic absorption band of pure PPy appears at around 930 cm^{-1} , which is attributed to the C–H wagging vibrations of the pyrrole ring. In the composite samples, this band shifts slightly to lower wavenumbers 928 cm^{-1} for PPy–TiO₂ and 924 cm^{-1} for PPy–SnO₂ indicating possible interactions between the PPy chains and the metal oxide particles. [45] These shifts suggest changes in the molecular environment of the polymer due to composite formation. A distinct absorption band observed at approximately 1554 cm^{-1} in pure PPy and PPy–SnO₂, and at 1550 cm^{-1} in the PPy–TiO₂ composite, corresponds to the C=C stretching vibrations within the pyrrole ring. This band signifies the conjugated double bond system characteristic at the polymer backbone. Additionally, a weaker band near 1462 cm^{-1} in pure PPy, shifting to 1458 cm^{-1} in PPy–SnO₂ and 1463 cm^{-1} in PPy–TiO₂, is attributed to C–C stretching vibrations in the pyrrole ring. [45] These slight shifts in wavenumber suggest subtle changes in the molecular interactions and structural conformation of the polymer matrix upon composite formation. The peaks observed in composite samples obtained a $1317\text{--}1334 \text{ cm}^{-1}$ can be attributed to C–N bonding in plane deformation. [45] The C–C stretching vibrations are identified at 1207 cm^{-1} in pure polypyrrole (PPy) and slightly shifted to 1203 cm^{-1} in the PPy–TiO₂ composite. A distinct band appearing at 1049 cm^{-1} corresponds to the C–H in-plane deformation mode of PPy. Furthermore, the presence of characteristic peaks at 794 cm^{-1} in both PPy and PPy–TiO₂, along with a peak at 777 cm^{-1} observed in the PPy–SnO₂ composite, indicates successful polymerization of the pyrrole monomer. In the PPy–TiO₂ composite, a broad absorption band near 3997 cm^{-1} is attributed to the stretching vibrations of hydroxyl (–OH) groups. The band around 1614 cm^{-1} corresponds to the bending vibration of Ti–OH, indicating adsorbed water on the TiO₂ surface. In the lower frequency region, bands between 450 and 750 cm^{-1} are associated with Ti–O–Ti symmetric stretching and Ti–O bond vibrations. Additionally, peaks observed in the range of $2000\text{--}2500 \text{ cm}^{-1}$, particularly around 2360 cm^{-1} , are linked to N=C=O bonding and suggest the formation of Ti–O–C=N and Ti–O=C–N linkages, resulting from the interaction between PPy and TiO₂. [46]

The analysis peaks of PPy–TiO₂–SnO₂ in Fig. 4, showed both the absorption peaks of TiO₂ and SnO₂ as it is discussed above separately for PPy–TiO₂ and PPy–SnO₂ composites. The amalgamation of both metal oxide peaks in PPy–TiO₂–SnO₂ evident of the formation of tri-composite material.

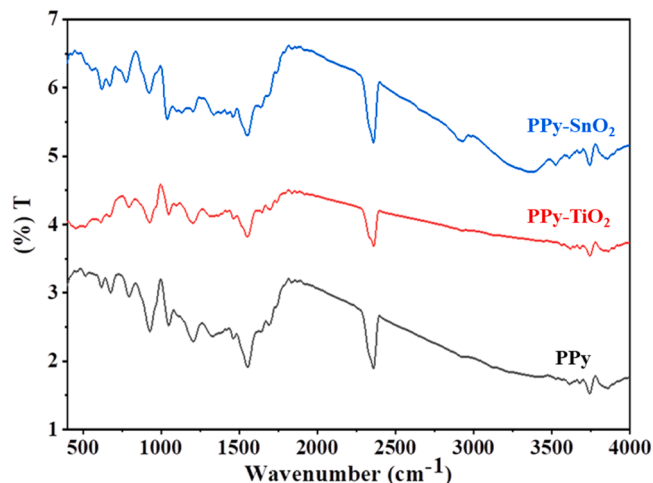


Fig. 3. FTIR spectra for Pure PPy, PPy–TiO₂, PPy–SnO₂ Nano-composites.

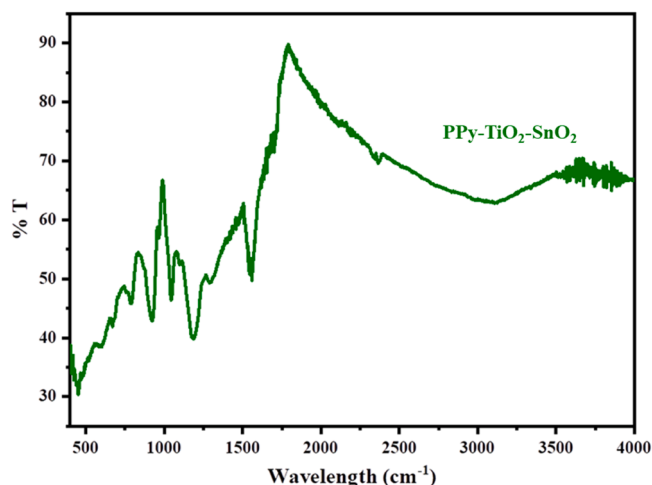


Fig. 4. FTIR spectra for PPy-TiO₂-SnO₂ tricomposite.

3.3. UV-Visible spectroscopic analysis

The UV-Visible absorption spectra of polypyrrole (PPy) and its nanocomposite thin films were recorded in the wavelength range of 275–450 nm at room temperature as depicted in Fig. 5(a).

The films were deposited using screen printing technique on thoroughly cleaned glass plates (slides). A cleaned, uncoated glass plate was used as the reference during the spectral measurements to ensure accurate baseline correction. A holmium oxide glass filter is used for wavelength calibration and validation of UV spectrophotometer for measuring the absorption spectra. [47] The samples were prepared using a binder solution composed of N-butyl carbitol and ethyl cellulose mixed in a fixed ratio. For each sample, the appropriate mass was accurately weighed and mixed with the binder solution in a 1:4 ratio (sample to binder) to obtain a uniform coating mixture. The absorbance spectrum of the sample was recorded by subtracting the reference signal from the sample reading. During the entire process, proper safety precautions were followed, including the use of facial protection and gloves. The work was carried out in a well-ventilated area, avoiding any direct sunlight, heated surfaces, or ignition sources. Furthermore, the optical band gap of Polypyrrole composite with TiO₂, SnO₂ and TiO₂-SnO₂ together has been determined using Tauc's equation given by $(\alpha h\nu) = A(h\nu - E_g)^m$. In this study, the proportionality constant is denoted by

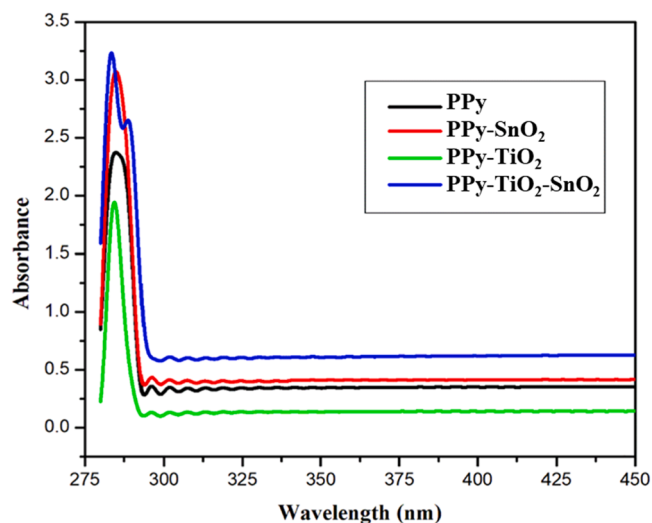


Fig. 5a. UV-Visible spectra of pristine Polypyrrole, PPy-SnO₂, PPy-TiO₂, PPy-TiO₂-SnO₂.

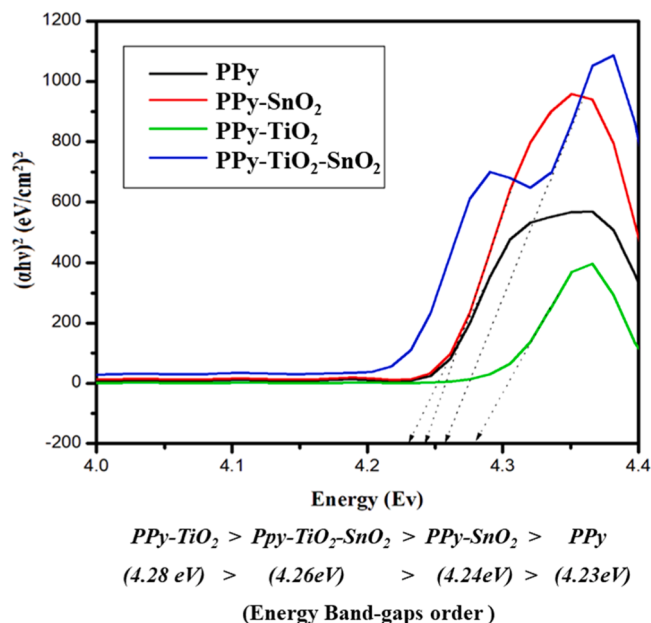


Fig. 5b. UV-Visible Tauc's plots of pristine Polypyrrole, PPy-SnO₂, PPy-TiO₂, PPy-TiO₂-SnO₂ and their corresponding optical energy band gaps.

'A', with 'E_g' representing the optical band gap energy, 'hν' the photon energy, and 'm' the transition factor. For direct allowed electronic transitions, the value of 'm' is taken as 1/2, as commonly used in literature. [48] The optical band gap of each sample was determined by plotting $(\alpha h\nu)^2$ against $h\nu$ and identifying the energy value at which the linear portion of the curve intersects the energy axis upon extrapolation as shown in Fig 5(b). [49,50]

The PPy and its composites showed the characteristic π to π^* electronic transition peak at 287–289 nm signifies the formation of conjugated structure of PPy. [51] The slight shifting in absorbance at wavelengths had occurred due to metal oxide doping resulting in variations of band gaps particularly in increasing trend. [52] The electronic band gap of chemically polymerized pure PPy sample is found to be 4.23 eV. It is found that the 10% wt/V doping caused band gap to increase for all the samples. Band gaps of PPy-SnO₂ and PPy-TiO₂ is found to be 4.26 eV and 4.28 eV while the band of tricomposite material PPy-TiO₂-SnO₂ lies between both the composite materials being 4.26 eV making it a novel material for gas sensing.

3.4. Proposed material engineering and sensing mechanism in Novel PPy-TiO₂-SnO₂

The work function is simply defined as the minimum energy required removing an electron from a solid to a point immediately outside the solid surface. When a composite is formed between TiO₂ or SnO₂ with the conducting polymer Polypyrrole (PPy), the work function of these metal oxide is modified due to charge transfer and interface effects. [53] Polypyrrole may introduce surface functional groups like oxygen-containing groups, which further shift the work function of the metal oxides TiO₂ and SnO₂. [54] In our study of EDAX data we found the significant percentage of the ions like Cl⁻, SO₄²⁻ in PPy which upon doping have further modified the Fermi levels of the composites and have influenced electronic properties of PPy-SnO₂ and PPy-TiO₂ nanocomposites. Also the p-type polypyrrole enhances hole accumulation at the interface, increasing the work function of the metaloxide composites. [55]

Thus when PPy- TiO₂ and PPy-SnO₂ were grounded together the pinning of their individual Fermi levels takes place and new material PPy-TiO₂-SnO₂ have found mid Fermi level lying between two metal

oxide composites. We obtain the energy band gap of this material practically as 4.26 eV which smoothly lies between the band gap energies of PPy-TiO₂ and PPy-SnO₂, thus forming an excellent material for gas sensing purpose.

Thus in novel PPy-TiO₂-SnO₂ tri-composite we got two n-type materials PPy-TiO₂ and PPy-SnO₂ whose energy band gaps were refined for producing vacancies at their micro-interfaces. Both tin dioxide (SnO₂) and titanium dioxide (TiO₂) blended with Polypyrrole are found to be wide band gap n-type semiconductors. [56] The direct energy band gap for PPy-TiO₂ composite material is found to be 4.28 eV while PPy-SnO₂ band gap is found to 4.24 eV. For the p-n type heterojunction, its sensing mechanism is equivalent to the loaded noble metal's electronic sensitization. Thus it is evident that when embedded into the Polypyrrole matrix it is plausible and noteworthy that the work function for PPy-TiO₂ becomes lower as compared to the work function for PPy-SnO₂. Thus Fermi energy level of PPy-TiO₂ is higher than that of PPy-SnO₂ because of its smaller work function so electron transfer occurs from the conduction band of PPy-TiO₂ to the conduction band of PPy-SnO₂ as both are embedded inside same conducting Polypyrrole matrix. [57]

This difference in work function between the TiO₂ and other n-type SnO₂ sensing material will induce the directional transfer of electrons from a lower work function to a higher one via balancing of the Fermi levels when they come into contact with each other. [58] As a result, an electron depletion layer on the material with low work function and an electron accumulation layer on the material with higher work function are formed. [59] It is anticipated that the electron accumulation layer may have facilitated the adsorption of carbon dioxide on the surface of the sensing materials. When the sensor is exposed to ambient air, oxygen molecules from the atmosphere adsorb onto the surface of the composite and capture electrons from its conduction band resulting in the formation of negatively charged oxygen species such as O₂⁻ and O⁻. [60] Consequently a stable state is established initially to serve as a baseline for the sensor. When CO₂ is introduced, it interacts with the pre-adsorbed oxygen species [61] and further traps the available electrons from material surface thereby again expanding the depletion layer within PPy-TiO₂-SnO₂ which leads to increase in surface resistance. Consequently, this innovative composite demonstrates a superior sensing response compared to alternative materials, as the microsites formed between TiO₂ and SnO₂ create active adsorption sites and surface oxygen chemistry crucial for gas interaction. In addition, PPy contributes a flexible conductive network that effectively translates interfacial charge disturbances into a distinct electrical signal. Fig. 6 represents the energy band diagram of PPy-TiO₂ and PPy-SnO₂ n-n type heterojunction at the interface. This sensing behaviour is consistent with recent studies

on advanced hybrid gas sensors based on PPy, where the formation of heterojunction interfaces and modulation of electronic structure significantly influence gas adsorption and charge transport mechanisms. [62] It has been reported that multi-component composite systems enhance sensing response through synergistic effects such as improved carrier mobility, increased active sites like inversion hole layer formation accelerating chemical reactions on the sensing materials surface. [63] The present PPy-TiO₂-SnO₂ system follows a similar trend, where dual heterointerfaces and modified band alignment contribute to improved sensing performance.

3.5. SEM Analysis

The microstructure profoundly affects the sensing characteristics of metal oxide-polymer composites. The surface morphology of the pure Polypyrrole and its composites with TiO₂ and SnO₂ were analysed using SEM and the pictures are as shown in Figs. 7(a),7(b),7(c). Microstructure sample images were captured at a magnification scale of 1 μ m for each respective sample. The amorphous characteristics observed in the SEM images are also evident in the XRD spectra of the sample composites.

The surface morphology of the synthesized pure polypyrrole (PPy) micrograph Fig. 7(a) reveals an irregular granular morphology with agglomerated spherical particles distributed throughout the surface. The particle boundaries are clearly distinguishable, indicating the formation of a porous, cauliflower-like structure, which is typical of chemically polymerized PPy. [64] Particle size analysis was carried out by measuring several randomly selected particles across the micrograph. The particle sizes we saw were between 187.8 nm and 222.3 nm, with an average diameter of about 203.65 nm. The consistent particle size distribution indicates that homogeneous nucleation occurred during the oxidative polymerization process. [65] Such nanoscale morphology is advantageous as it enhances the effective surface area, making the material promising for applications in sensor devices. [66] Both smaller and larger voids were observed in the morphology of the pure PPy micrograph.

The SEM micrograph images 7(b), 8(a) of the PPy-TiO₂ composite shows a granular morphology with well-connected spherical particles forming dense clusters. In comparison to pure PPy, the addition of TiO₂ seems to enhance the compactness of the agglomerated grains, leading to a more interconnected network. The particle size distribution histogram confirms this observation, showing a relatively narrow distribution with grain sizes ranging from 195 to 225 nm. The calculated average grain size is 211.36 nm, which is slightly larger than that of

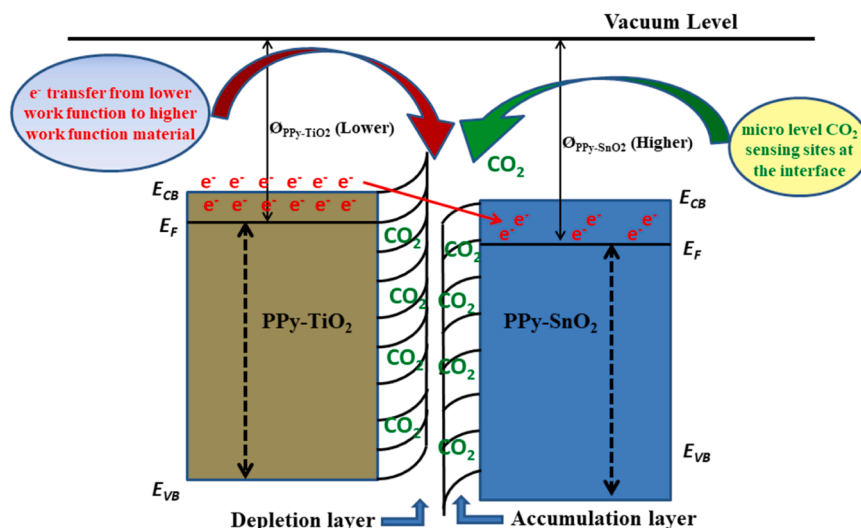


Fig. 6. Proposed schematics of electronic transition and micro CO₂ sensing sites in novel PPy-TiO₂-SnO₂ tri-composite.

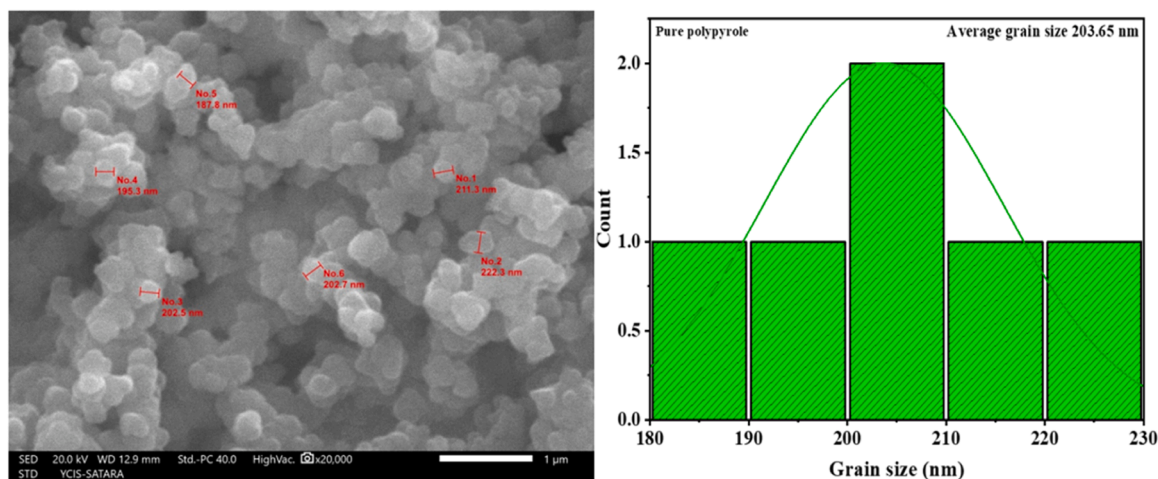


Fig. 7a. SEM image of the Pure Polypyrrole (PPy) sample with its particle size distribution histogram.

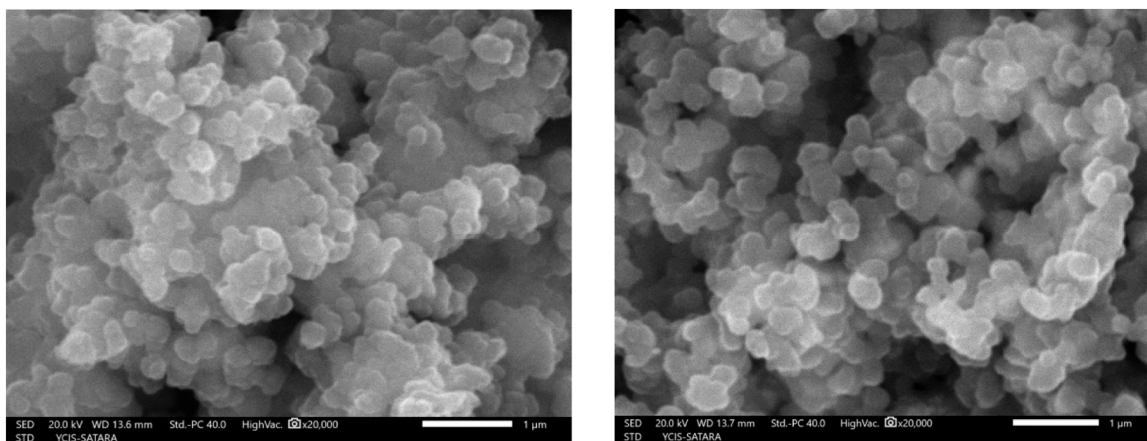


Fig. 7b. High-resolution SEM images of the PPy-TiO₂ and 7(c) PPy-SnO₂ samples captured at a magnification scale of 1 μm.

pristine PPy (203.65 nm). This moderate increase in grain size can be attributed to the uniform embedding of TiO₂ nanoparticles within the PPy matrix, which acts as nucleation centres and facilitates controlled growth. [67]

The SEM image of the PPy-SnO₂ composite Fig. 7(c), 8(b) displays a similar granular morphology but with noticeably larger and more distinct spherical grains compared to both pure PPy and the TiO₂-doped sample. The particle boundaries are sharper, and the overall structure appears less compact, suggesting that SnO₂ incorporation influences the growth kinetics of PPy during polymerization. [68] The particle size distribution histogram shows grain sizes in the range of 205–238 nm, with an average value of 220.18 nm, which is higher than both pure PPy and the PPy-TiO₂ composite. The relatively broader size distribution indicates that SnO₂ particles promote heterogeneous nucleation, leading to slightly coarser grain formation. [69] Such an increase in grain size, coupled with the retained porous morphology, may enhance ion diffusion and improve electrochemical accessibility, making the PPy-SnO₂ composite a strong candidate for electrochemical and sensing applications.

The integration of TiO₂ and SnO₂ within the PPy matrix provides a synergistic effect, wherein TiO₂ contributes enhanced stability and photocatalytic activity, while SnO₂ ensures superior electron mobility and gas-sensing response. [70] The coexistence of these wide band gap semiconductors with conductive PPy facilitates efficient charge transfer, improved interfacial interactions, and a higher density of active adsorption sites shown in our proposed diagram Fig. 8.

3.6. Elemental analysis

The Energy Dispersive X-ray spectra of Pristine PPy, PPy-TiO₂, and PPy-SnO₂ were recorded as shown in Fig. 9 and the corresponding elemental compositions were systematically presented in tabulated form (Table 2).

The EDX spectrum of pristine PPy confirms the expected dominance of carbon (54.11 wt%, 59.89 at%) and nitrogen (40.00 wt%, 37.96 at%), consistent with the conjugated pyrrole backbone. The presence of chlorine (5.48 wt%, 2.06 at%) and a trace of iron (0.40 wt%, 0.10 at%) is observed; these signals are attributed to residual FeCl₃, the oxidizing agent used during chemical polymerization. The low iron content indicates that Fe residues are minimal after washing, while the chlorine peak reflects chloride counter-ions associated with the doped polymer. [71] Overall, the elemental profile confirms successful synthesis of PPy with expected dopant residues from FeCl₃. The PPy-TiO₂ composite exhibits significant oxygen (26.12 wt%, 23.07 at%) and measurable titanium (4.14 wt%, 1.22 at%) signals in addition to the PPy matrix elements (C, N). Chlorine and iron traces are again present at low levels, consistent with residual FeCl₃ from the polymerization step. The combined presence of Ti and O confirms TiO₂ incorporation into the polymer matrix. [72] The identified Fe and Cl are not inherent to the oxide but rather stem from the oxidant and their minimal quantities suggest efficient but incomplete elimination of them during sample washing. The elemental composition therefore validates the formation of the PPy-TiO₂ nanocomposite and supports the morphological observations

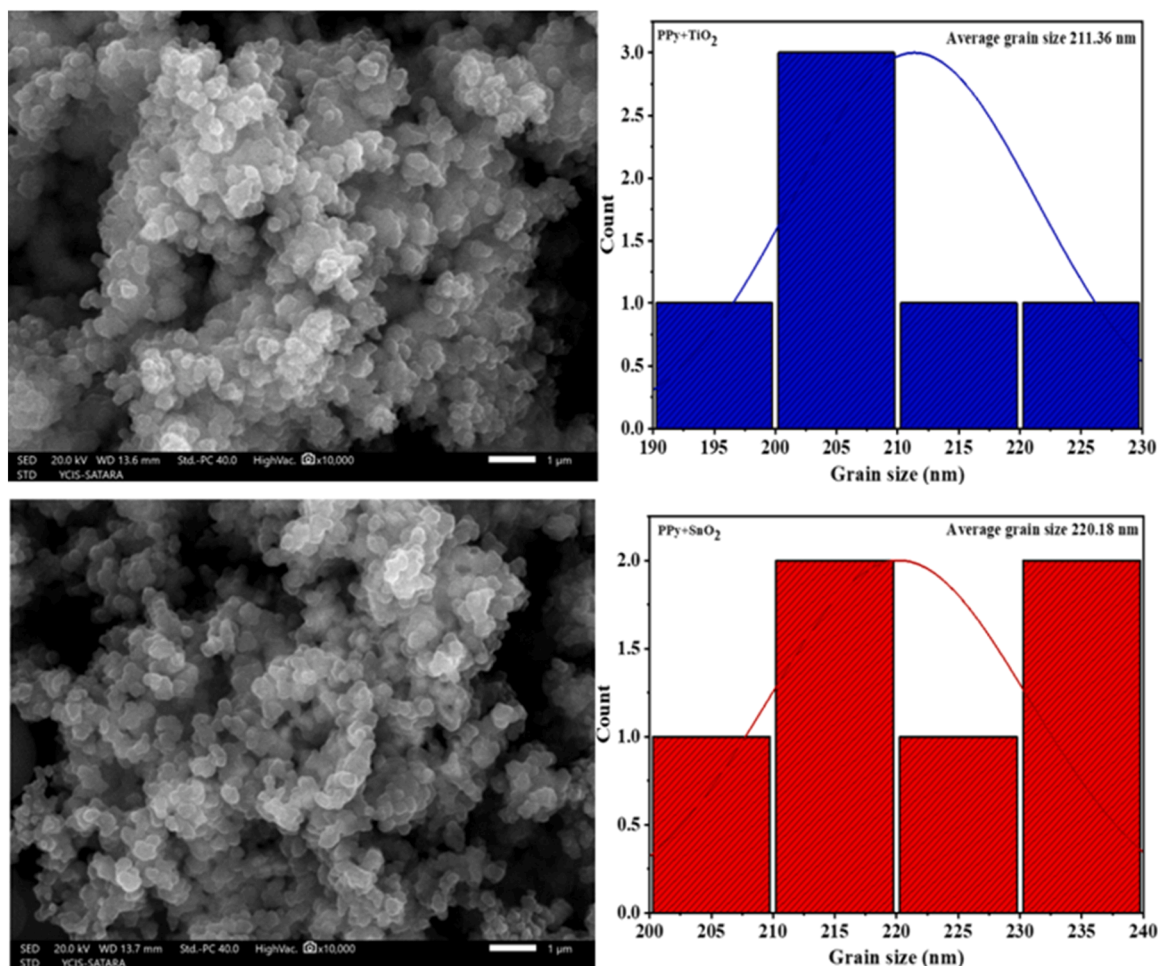


Fig. 8. (a) and 8(b): SEM images of the PPy-TiO₂ and PPy-SnO₂ samples showing well-connected granular surface morphology along with their respective histogram depicting average grain size.

from SEM. The EDX spectrum for the PPy-SnO₂ sample shows clear peaks corresponding to tin (Sn) and oxygen, confirming the presence of SnO₂ within the PPy matrix. Carbon and nitrogen remain the dominant signals from the polymer. As with the other samples, small amounts of chlorine and iron are detected and are ascribed to residual FeCl₃ from the oxidative polymerization. The coexistence of conducting PPy with semiconducting SnO₂, as corroborated by EDX, suggests the formation of active PPy-SnO₂ interfaces that are expected to promote interfacial charge transfer and improve sensor response. [73]

3.7. Thermo Gravimetric Analysis

Thermogravimetric analysis (TGA) of pristine polypyrrole (PPy), PPy-TiO₂ (10 wt%), PPy-SnO₂ (10 wt%), and the ternary PPy-TiO₂-SnO₂ composite was performed in the range of room temperature to 600 °C under a nitrogen atmosphere (Fig. 10). All samples exhibited an initial mass loss below 150 °C, which can be attributed to the evaporation of physically adsorbed moisture and removal of volatile methanol remaining from the synthesis process.

Here the pristine PPy displayed two-step degradation behaviour. The first step occurred between 150–250 °C, corresponding to the release of dopants and low-molecular-weight fragments [74], while the second major weight loss region, above 300 °C, was assigned to backbone degradation and partial carbonization. A considerable char residue was retained at 600 °C, consistent with the typical thermal behaviour of conducting polymers. [75]

In contrast, the binary composites PPy-TiO₂ and PPy-SnO₂ showed

accelerated degradation compared to pristine PPy, as evidenced by sharper weight losses and reduced residual masses at 600 °C. This suggests that the oxide nanoparticles promote polymer chain scission and volatile release, possibly due to their catalytic influence on the degradation process. [76,77]

Interestingly, the ternary composite PPy-TiO₂-SnO₂ demonstrated markedly improved thermal stability. Only negligible mass loss was observed up to 600 °C, which can be attributed to the higher inorganic fraction and possible synergistic effects between SnO₂ and TiO₂. These factors likely restricted chain mobility and hindered thermal decomposition of the PPy matrix. Here the TGA results indicate that while incorporation of individual oxides tends to reduce the thermal stability of PPy but the ternary system exhibits enhanced resistance to decomposition, making it suitable for applications requiring high-temperature durability in Sensors.

3.8. Impedance Analysis

Impedimetric analysis was investigated using an impedance analyzer. The electrical impedance spectra of PPy, PPy-SnO₂, PPy-TiO₂, and PPy-TiO₂-SnO₂ were recorded over a frequency range of 269 Hz –5 MHz using a HIOKI impedance analyzer. The impedance spectroscopy was used in various applications such as solar cell [78], supercapacitors [79], biosensors [80] and fuel cells [81]. The complex impedance was analyzed in terms of its real (Z') and imaginary (Z'') components, as depicted in Fig. 11(a).

The Nyquist plot reveals the presence of a non-Debye type relaxation,

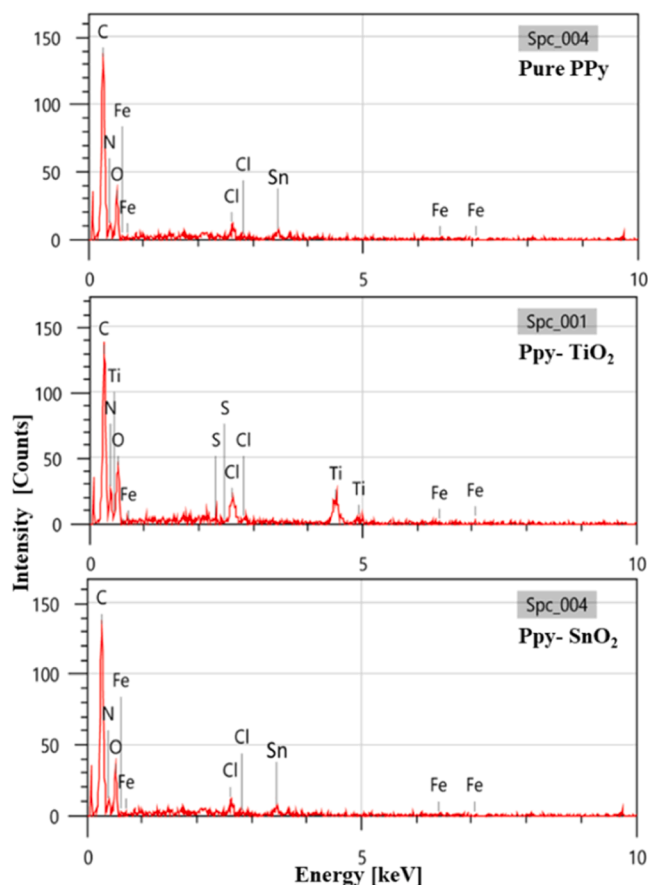


Fig. 9. EDAX scan for Pure PPy, PPy-TiO₂ and PPy-SnO₂.

Table 2

Elemental mass and atomic percentage of Pure PPy, PPy-TiO₂ and PPy-SnO₂ nanocomposites.

Element	Line	Mass %	Atom %
C	K	54.11	59.89
N	K	40.00	37.96
Cl	K	5.48	2.06
Fe	K	0.40	0.10
Total		100.00	100.00
Element	Line	Mass %	Atom %
C	K	42.23	49.68
N	K	24.74	24.96
O	K	26.12	23.07
S	K	0.40	0.18
Cl	K	2.05	0.82
Ti	K	4.14	1.22
Fe	K	0.32	0.08
Total		100.00	100.00
Element	Line	Mass %	Atom %
C	K	48.1	56.5
N	K	18.6	18.8
O	K	26.4	23.2
Cl	K	1.27	0.5
Fe	K	0.20	0.05
Sn	L	7.3	0.9
Total		100	100

characterized by a lack of a typical semicircular arc. This observation aligns with findings in several literature reviews, which also link non-Debye relaxation to a depressed semi-circular curve. [82,83] Additionally, Fig. 11(b) illustrates the complex impedance spectra derived from circuit analysis. The impedance spectra were fitted, and electrical

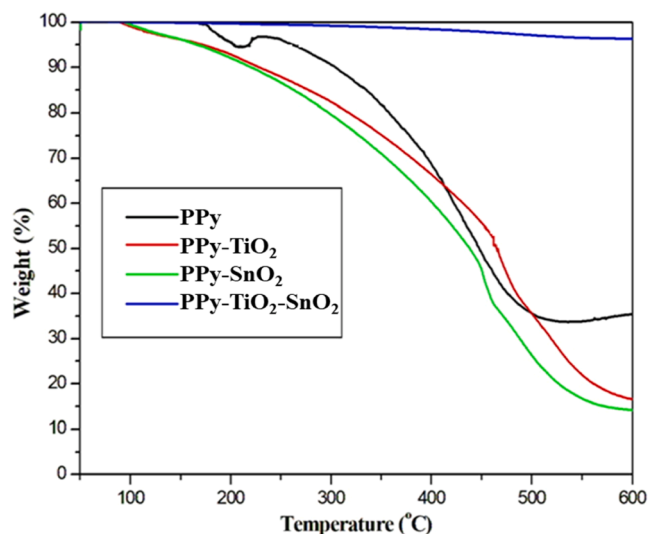


Fig. 10. Thermogravimetric analysis (TGA) of pristine polypyrrole (PPy), PPy-TiO₂ (10 wt%), PPy-SnO₂ (10 wt%), and the ternary PPy-TiO₂-SnO₂ composite.

parameters were obtained using the Z-sim software, as detailed in Table 3. The electric circuit exhibits R-(CR)-W type characteristics, attributed to the inherent properties of the polymer, and this setup allows for the calculation of relaxation times to better understand the conduction process. The resistances R, R_{dc}, and R_{ct} represent high-frequency impedance, low-frequency impedance, and surface resistance, respectively. The presence of the Warburg constant (W) also confirms the linear nature of complex impedance spectra at 45°. Among the samples, PPy-TiO₂ displayed a higher Warburg constant, which indicates reduced conductivity—a finding that is verified by its optical energy band gap, the largest among the samples. This correlation was consistently observed across all samples.

3.9. CO₂ gas sensing application

Gas sensing response is obtained in an ambient CO₂ environment, where the concentration of the gas is varied from the lowest ppm (parts per million) level to higher values. The gas sensing response is characterized using three primary types of responses as linear response, static response, and dynamic response.

The change in resistance of the sensor in presence of gas leads to the change in sensitivity which is well described by a formula [84],

$$\text{Sensitivity (S)} = \frac{R_g - R_a}{R_a} = \frac{\Delta R}{R_a}$$

Where, R_a = Resistance in presence of air, R_g = Resistance in presence of gas

The resistance of all the sample sensors were found to be increasing with increase in the concentration of CO₂ gas inside the chamber. The linear sensitivity response of all composite sensors was obtained by continuously passing the CO₂ gas inside a chamber for 4 min at 1 Litre/min at constant pressure. The increase in gas concentration inside the chamber leads to more adsorbance of CO₂ molecules on the sensor surface leading to the increase of resistance of sensor as compared inside the presence of air. A graph of sensitivity with respect to the time of passing of CO₂ gas is plotted to obtain the linear responses of various composites. The linear response so obtained is mentioned in graph (Fig. 12) and highest sensitivities at 4 min exposure of sensors to the carbon dioxide gas are depicted in Table 4.

It has been observed that the pure polypyrrole sample and its composite with titania nanoparticles exhibited higher sensitivities of 12.38

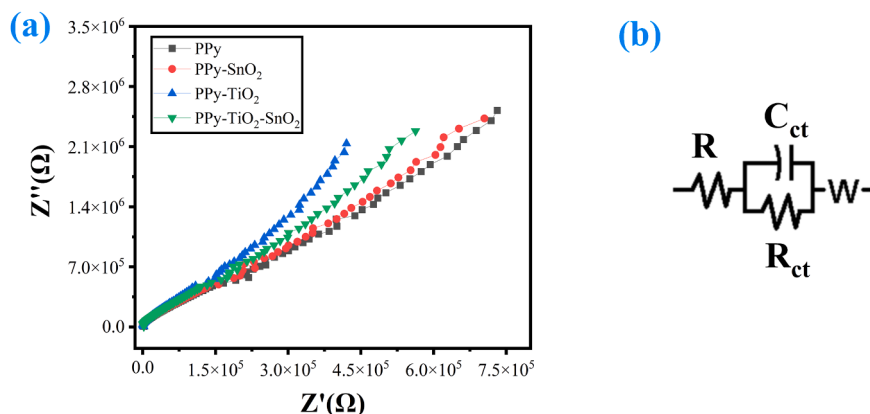


Fig. 11. (a) Nyquist plot of impedance for samples PPy, PPy-SnO₂, PPy-TiO₂, and PPy-TiO₂-SnO₂, and (b) The impedimetric equivalent circuit R(CR)W for all the samples.

Table 3

The impedimetric equivalent circuit parameters for polypyrrole matrix.

Samples	R	R _{ct}	C _{ct}	τ _{ct}	R _{dc} (K Ω)	C _{dc} (μF)	T _{dc} (ms)	W (μ Ω. s ^{0.5})
PPy	0.147 (m Ω)	2.263 (m Ω)	0.0582 (pF)	0.13 (fs)	15.83	1	15.83	0.01251
PPy-SnO ₂ 10%	0.65 (m Ω)	1.51 (m Ω)	0.51 (pF)	0.77 (fs)	15.46	1	15.46	0.01281
PPy-TiO ₂ 10%	37.35 (m Ω)	361 (K Ω)	0.44 (μF)	158.84 (ms)	11.78	1	11.78	0.0137
PPy-TiO ₂ -SnO ₂ 10%	14.6 (Ω)	4.95 (KΩ)	0.0159 (μF)	78.71 (μF)	13.62	1	13.62	0.0133

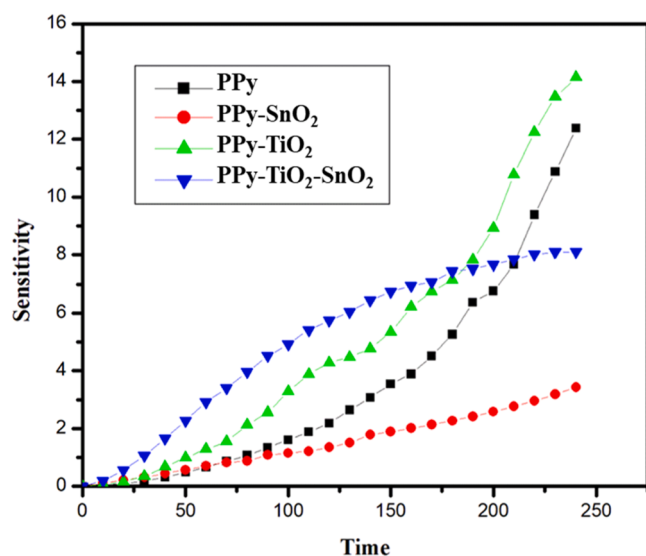


Fig. 12. Linear response of gas sensors taken at 4 min.

Table 4

Maximum Linear sensitivity response of gas sensors taken for 4 min.

Nanocomposite	Max. Sensitivity
PPy	12.38
PPy-SnO ₂	3.43
PPy-TiO ₂	14.15
PPy-TiO ₂ -SnO ₂	8.09

and 14.15 respectively over other composites. It has been observed that

the maximum sensitivity response of novel PPy-TiO₂-SnO₂ tricomposite is equal to 8.09 and suitably lies between the sensitivity response of PPy-TiO₂ and PPy-SnO₂ giving an insightful evidence of alteration of energy band gaps while forming a tricomposite sensor.

3.9.1. Static Response of gas sensors

The static response represented the steady-state behaviour of the gas sensors when exposed to a fixed concentration of CO₂ in parts per million for a prolonged period taken at various lower concentrations of 50 ppm, 100 ppm, and 150 ppm is as shown in Fig. 13(a), 13(b), 13(c) respectively. Although minor fluctuations are observed, each response curve at different ppm levels clearly exhibits an initial stable baseline, a quasi-

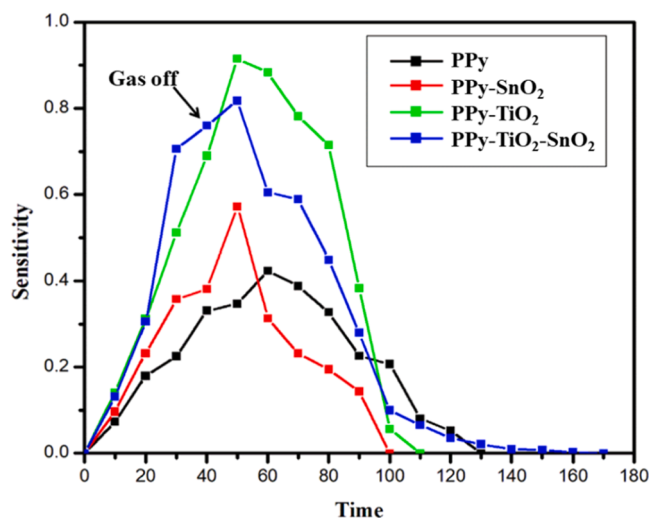


Fig. 13a. Static Response of fabricated composite sensors over CO₂ gas at 50 ppm.

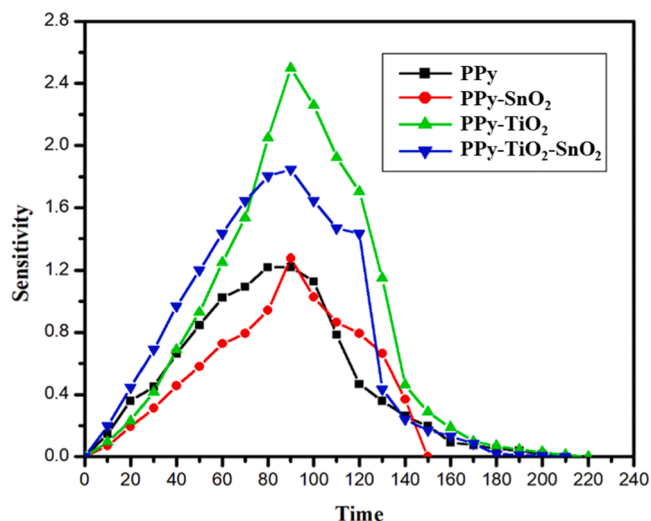


Fig. 13b. Static Response of fabricated composite sensors over CO₂ gas at 100 ppm.

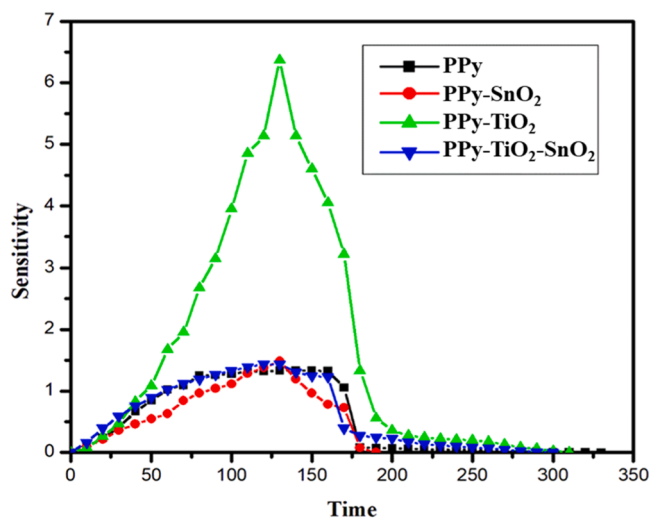


Fig. 13c. Static Response of fabricated composite sensors over CO₂ gas at 150 ppm.

steady resistance plateau upon CO₂ exposure, and a recovery trend toward the baseline after gas removal which satisfies the criteria for reliable gas-sensing measurements.

All sensors responded well to the gas flow as soon as the pressure gauge was opened indicating the excellent method of synthesis followed in this work in developing the sensors. As the concentration of CO₂ increases, the response of each sensor progressively improves. For a concentration of 50 ppm, carbon dioxide (CO₂) was introduced into the chamber for 40 s. Subsequently, the gas flow was halted for an additional 40 s to assess the system's stability. After 40 s of stopping the CO₂ flow, the gas was rapidly removed by extracting the sensor from the chamber and exposing it to the ambient air. The same procedure was applied for CO₂ concentrations of 100 ppm and 150 ppm, with initial gas exposure durations of 80 s and 120 s, respectively, in order to obtain the corresponding static responses. In each case, the sensor response reached equilibrium upon stopping of the CO₂ flow giving small output signal of stability. After the gas was removed and the sensors were exposed to fresh air, all sensors fully returned to their original baseline values, showing 100% recovery of the response. The long term stability was shown by the novel material PPy-TiO₂-SnO₂ tricomposite than that

of other sensors at all ppm concentrations. The maximum sensitivity is shown by PPy-TiO₂ as it has highest energy band gap of 4.28 eV while pure PPy shows least sensitivity at the 50 ppm of low concentration of the gas. PPy-TiO₂-SnO₂ nanocomposite showed maximum intensity of 0.82 at 50 ppm as its band gap varies between the PPy-TiO₂ and PPy-SnO₂ nanocomposites.

It has been observed from static responses taken at 150 ppm of CO₂ evident that, when gas was stopped inside the chamber the novel PPy-TiO₂-SnO₂ tricomposite sensor maintained its stability without changing its sensitivity, ensuring the highest stability over other composites. Although the PPy-TiO₂ sensor exhibited the highest sensitivities across all tested concentrations (50, 100, and 150 ppm), it demonstrated limited stability at its maximum sensitivity levels when the gas was turned off. In contrast, the PPy-TiO₂-SnO₂ sensor showed more consistent and stable performance under the same conditions, indicating superior recovery characteristics and enhanced reliability which is well evident from the graphs. The energy band gap alteration and its effect on the sensitivity depicted in Table 5, shows that the obtained PPy-TiO₂ composite has highest sensitivities of 0.91, 2.50 and 6.36 at 50, 100, and 150 ppm level of CO₂ gas as the composite electronic band gap is found to be 4.28 eV which is measured highest, making it most sensitive over other sensors.

The sensing repeatability and reversibility of the PPy-TiO₂-SnO₂ to 50- and 100-ppm CO₂ are shown in Fig. 14(a) and 14(b), respectively. The response value of the gas sensor returned to the original response value after exposure to 50- and 100-ppm CO₂ gas in the same time in each repetitive cycle respectively. The response and recovery behaviour over five continuous cycles indicates that the PPy-TiO₂-SnO₂ sensor exhibits good reproducibility, demonstrating that the sensor has reliable and consistent performance showing good repeatability. It is notable that the PPy-TiO₂-SnO₂ tri-composite sensor has excellent CO₂-gas sensing properties at room temperature and this suggests that it is an excellent candidate material for CO₂ gas detection.

From Fig. 14(a), 14(b) and 14(c) it is concluded that PPy-SnO₂-TiO₂ not only exhibit good repeatability but also shows good stability when gas was turned off. It has been seen from both graphs obtained at 50 ppm and 100 ppm that the novel sensor is very sensitive to CO₂ adsorption at its surface as the sensitivity falls significantly after stopping of the gas. To analyse the recovery time and response time more significantly we have studied the dynamic response of pure PPy, PPy-TiO₂, PPy-SnO₂, and PPy-SnO₂-TiO₂ with increasing ppm level of CO₂ inside gas chamber by 50 ppm.

3.9.2. Dynamic Response of gas sensors

To evaluate the dynamic response of the sensor, controlled concentrations of CO₂ gas at the rate of 1 L/s were introduced into a sealed metal chamber. Initially, CO₂ was injected into the chamber to achieve a concentration of 50 ppm, based on the known volume of the enclosure. The gas flow was maintained for 40 s, after which it was halted for an equal duration to allow stabilization. This cycle of gas injection and stoppage, each lasting 40 s, was repeated. With each successive cycle, the CO₂ concentration in the chamber increased cumulatively-reaching 100 ppm after the second cycle, 150 ppm after the third. This stepwise increase enabled the collection of dynamic response data for varying concentrations, as illustrated in Fig. 15.

Table 5

Highest sensitivities of obtained sensors at different lower concentrations of CO₂ gas.

Nano-Composite Sensors	Band Gaps (eV)	Sensitivity 50 ppm	Sensitivity 100 ppm	Sensitivity 150 ppm
PPy	4.23	0.42	1.22	1.33
PPy-SnO ₂	4.24	0.57	1.28	1.48
PPy-TiO ₂	4.28	0.91	2.50	6.36
PPy-TiO ₂ -SnO ₂	4.26	0.82	1.85	1.49

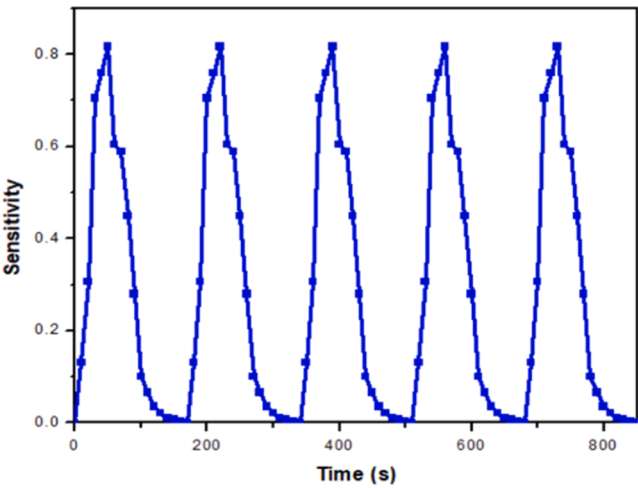


Fig. 14a. Sensing repeatability and reversibility of the PPy-SnO₂-TiO₂ at 50 ppm CO₂.

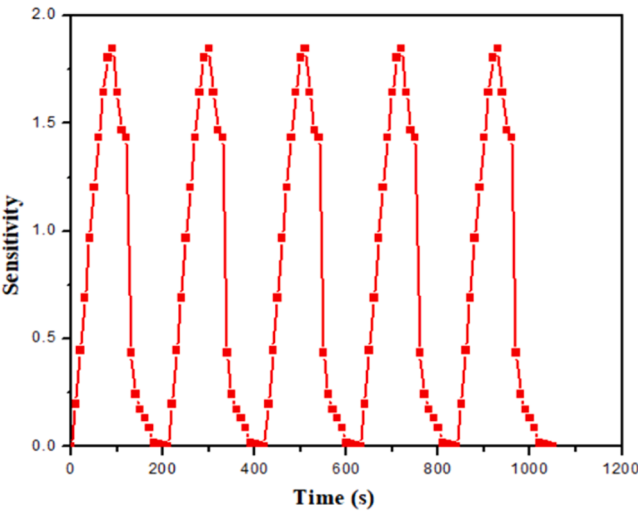


Fig. 14b. Sensing repeatability and reversibility of the PPy-SnO₂-TiO₂ at 100 ppm CO₂.

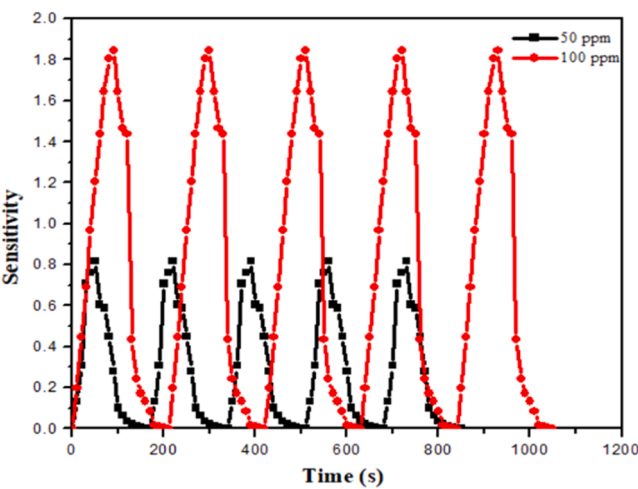


Fig. 14c. shows comparative sensing repeatability and reversibility of the PPy-SnO₂-TiO₂ at 50 and 100 ppm of CO₂ gas.

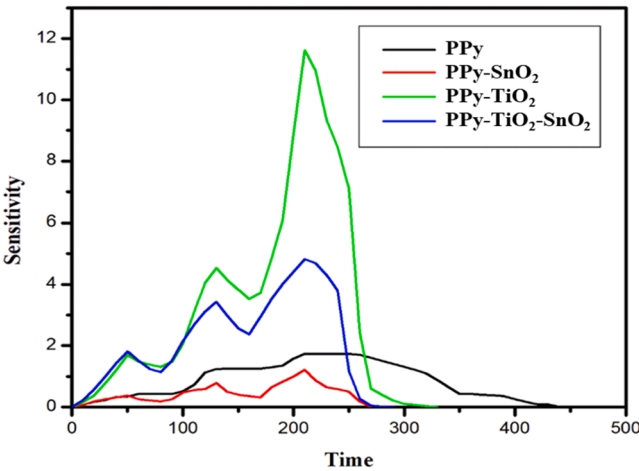


Fig. 15. Dynamic response of PPy, PPy-TiO₂, PPy-SnO₂, and novel PPy-TiO₂-SnO₂.

The response time is defined as the duration required for the sensor to reach 90% of its maximum response upon exposure to the target gas. Conversely, the recovery time refers to the duration the sensor takes to return to 90% of its baseline value after the removal of the gas. [85] Thus we required a sensor whose response and recovery time are less, the calculated response and recovery time for different composite sensors are depicted in Table 6.

Here the newly fabricated PPy-TiO₂-SnO₂ meets with an ability to produce short response time of just 184 s for higher ppm of the gas and also short response time of 76 s. This is achieved by overlapping the electronic properties of both metal oxides TiO₂ and SnO₂ which are made composite inside the Polypyrrole matrix. PPy-TiO₂ sensor showed higher sensitivity than other sensors but the response time to achieve maximum sensitivity is 184 s which is equal to the PPy-SnO₂-TiO₂ tri-composite. At the same time the recovery time of novel PPy-TiO₂-SnO₂ sensor is just 76 s which is very short as compared to other sensors making it most effective than others.

In PPy-TiO₂-SnO₂ nanocomposite sensor the interface between the depletion region created at PPy-SnO₂ with lower work function and the accumulation region created at PPy-TiO₂ provides the transition of electrons. Transition of electrons occurs from lower work function to higher work function and in this process every time it is plausible that voids may form due to repeated electron transitions at the interface inside the material at interfaces of depletion region and accumulation region. [86] This produces more vacancies, and hence more CO₂ molecules are absorbed, as compared to the conventional PPy-TiO₂ and PPy/SnO₂ composite materials thus. Hence, the obtained PPy-TiO₂-SnO₂ tri-nanocomposite sensor is the most efficient sensor from other fabricated sensors. PPy-TiO₂-SnO₂ sensor makes itself a novel material sensor working on very less gas concentrations upto 150 ppm present till now in studies.

Table 6
Response time and Recovery time for different synthesized composite sensors.

Nano-Composite Sensors	Response Time (s)	Recovery Time (s)
PPy	198 s	171 s
PPy-SnO ₂	189 s	54 s
PPy-TiO ₂	184 s	112 s
PPy-SnO ₂ -TiO ₂	184 s	76 s

3.9.3. Selectivity of PPy-TiO₂-SnO₂ sensor

To evaluate the selectivity of the developed sensor, the sensing response of the PPy-TiO₂-SnO₂ nanocomposite was investigated towards different gases, namely CO₂, NH₃, CO, and Ethanol, each at a fixed concentration of 50 ppm under identical experimental conditions. The corresponding selectivity responses are presented in Fig. 16.

It is clearly observed that the sensor exhibits the highest sensitivity towards CO₂ (0.82), followed by NH₃ (0.46), ethanol (0.35), and CO (0.27). This trend confirms the preferential sensing behaviour of the PPy-TiO₂-SnO₂ composite towards CO₂ gas.

The enhanced selectivity towards CO₂ can be explained based on the proposed sensing mechanism. CO₂, being a weakly acidic gas, interacts more effectively with electron-rich adsorption sites and pre-adsorbed oxygen species (O₂⁻, O⁻) present on the surfaces of TiO₂ and SnO₂, leading to modulation of charge carrier density and sensing response. [87] The presence of oxygen vacancies in SnO₂ further enhances CO₂ adsorption [88], while TiO₂ provides stable surface states facilitating gas interaction. In addition, the dual heterojunctions formed at PPy-TiO₂ and PPy-SnO₂ interfaces promote efficient charge redistribution and modulation of the depletion layer, resulting in a higher sensing response.

In contrast, NH₃, being a reducing gas, donates electrons to the sensing material and partially suppresses the depletion layer, resulting in a comparatively lower response. [89] Ethanol, a polar volatile organic compound, exhibits weaker surface interaction at room temperature, while CO shows minimal response due to its relatively low reactivity under ambient conditions. These results demonstrate that the synergistic effect of the conducting polymer matrix and metal oxide hetero-interfaces not only enhances the sensing performance but also imparts good selectivity towards CO₂ gas at room temperature.

3.9.4. Long-term stability and drift analysis of PPy-TiO₂-SnO₂ sensor

The long-term stability of the optimized PPy-TiO₂-SnO₂ sensor was evaluated to examine the retention of sensing performance over an extended period of time. The sensor response toward a fixed CO₂ concentration of 50 ppm was recorded on Day 0 and subsequently on Day 7, 14, 21, 28, and 35, while the sensor was stored under ambient laboratory conditions between measurements. The same experimental protocol was followed for each measurement to ensure consistency.

The variation in sensitivity as a function of time is presented in Fig. 17(a). It is observed that the sensor maintains a relatively stable response over the testing duration, with only a slight decrease from an initial value of 0.82–0.77 on Day 35. The calculated sensitivity drift was

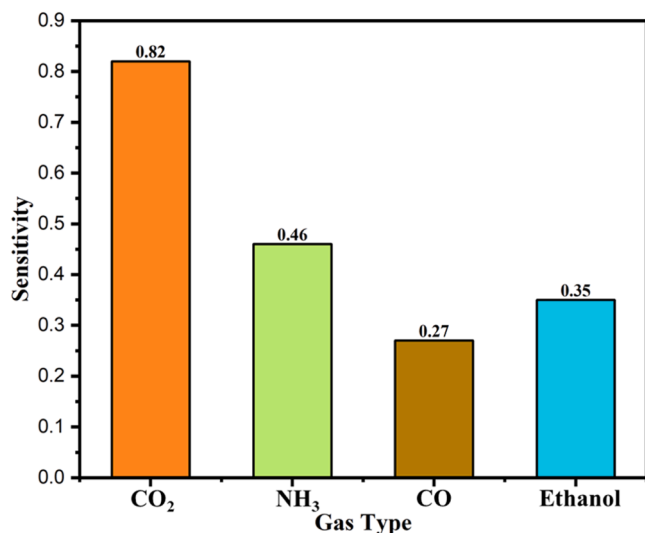


Fig. 16. Selectivity response of PPy-TiO₂-SnO₂ sensor towards CO₂, NH₃, CO, and ethanol at a fixed concentration of 50 ppm under identical experimental conditions.

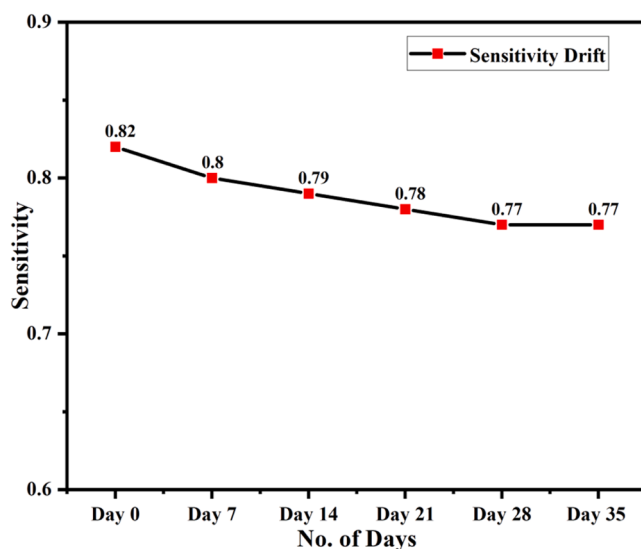


Fig. 17a. Variation of sensor sensitivity towards 50 ppm CO₂ as a function of time (days), showing sensitivity drift behavior of PPy-TiO₂-SnO₂ sensor.

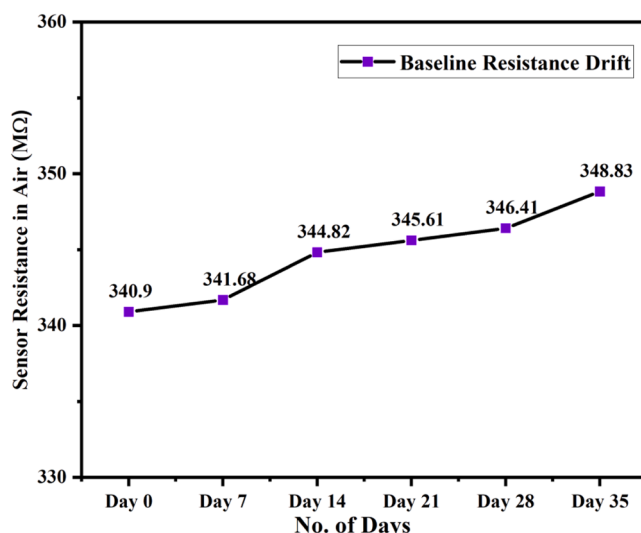


Fig. 17b. Baseline resistance variation of PPy-TiO₂-SnO₂ sensor in air over time (days), indicating baseline stability.

found to be 6.10%, indicating good retention of sensing performance over time.

The baseline stability of the sensor was also evaluated by monitoring the resistance in air, as shown in Fig. 17(b). The baseline resistance exhibited a marginal increase from 340.9 MΩ to 348.83 MΩ over 35 days, corresponding to a baseline drift of 2.33%. This slight increase in resistance may be attributed to gradual adsorption of ambient oxygen species and stabilization of the sensing surface. The detailed values of sensitivity, baseline resistance, and corresponding drift percentages are summarized in Table 7.

The relatively low sensitivity drift (<7%) and minimal baseline variation (<3%) demonstrate the good long-term stability and reliability of the PPy-TiO₂-SnO₂ sensor. The observed stability can be attributed to the synergistic effect of the composite structure, where TiO₂ and SnO₂ provide chemical and thermal robustness, while the PPy matrix ensures efficient charge transport and structural flexibility. These results confirm that the developed sensor is suitable for practical CO₂ sensing applications with stable long-term performance.

Table 7

Long-term stability and drift analysis of PPy–TiO₂–SnO₂ sensor at 50 ppm CO₂ over 35 days.

Days	Sensitivity (S _t)	Sensitivity Drift (%) = $\frac{(S_0 - S_t)}{S_0} \times 100$	Baseline Resistance (MΩ)	Baseline Drift (%) = $\frac{(R_t - R_0)}{R_0} \times 100$
Day 0	S ₀ = 0.82	0	340.9	R ₀ = 0
Day 7	0.8	2.44	341.68	0.23
Day 14	0.79	3.66	344.82	1.15
Day 21	0.78	4.88	345.61	1.38
Day 28	0.77	6.10	346.41	1.61
Day 35	0.77	6.10	348.83	2.33

3.9.5. Effect of relative humidity on sensing performance

The effect of relative humidity (RH) on the sensing performance of the PPy–TiO₂–SnO₂ sensor was investigated under controlled conditions. Humidity levels of approximately 30%, 50%, and 70% RH were achieved using saturated aqueous solutions of MgCl₂, Mg(NO₃)₂, and NaCl, respectively. For each condition, 100 ml of the respective saturated salt solution was placed in an open beaker inside the gas chamber and the system was allowed to stabilize for 60 min. All measurements were carried out at room temperature. After stabilization, CO₂ gas at a fixed concentration of 50 ppm was introduced into the chamber at a constant flow rate of 1 L/min, and the corresponding sensor response was recorded. The variation in sensitivity with RH is shown in Fig. 18.

It is observed that the sensor exhibits only a slight variation in response with increasing humidity, with sensitivity values changing marginally from 0.82 to 0.79. This indicates good humidity tolerance of the ternary PPy–TiO₂–SnO₂ sensor. The minor decrease in sensitivity at higher RH can be attributed to competitive adsorption of water molecules on active sites, which slightly reduces CO₂ adsorption. Overall, the results demonstrate that the developed sensor maintains stable performance under varying humidity conditions.

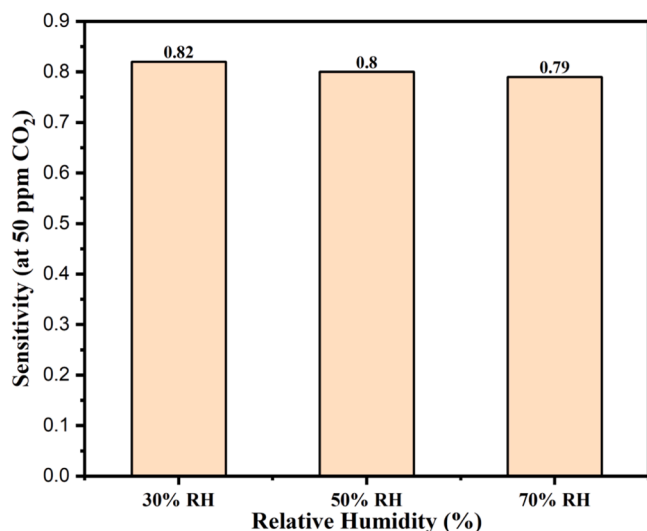


Fig. 18. Variation of sensitivity of PPy–TiO₂–SnO₂ sensor towards 50 ppm CO₂ at different relative humidity levels (30%, 50%, and 70% RH) at room temperature.

4. Conclusion

PPy–TiO₂, PPy–SnO₂, and innovative PPy–TiO₂–SnO₂ sensors were developed successfully using an in-situ chemical oxidative polymerisation technique, followed by screen printing the material onto chemically cleaned glass slides. X-ray diffraction analysis confirmed the successful formation of structurally coherent PPy–TiO₂, PPy–SnO₂, and PPy–TiO₂–SnO₂ nanocomposites, where the crystalline oxide phases remain preserved within the amorphous PPy matrix. The tricomposite displayed the characteristic reflections of both TiO₂ (anatase/rutile) and SnO₂ (cassiterite), superimposed on the broad halo of PPy, demonstrating the coexistence of crystalline semiconducting oxides with an amorphous conducting polymer. The crystallite size extracted using the Debye–Scherrer relation exhibited a progressive enhancement from 12.34 nm (TiO₂) and 20.52 nm (SnO₂) to 35.97 nm in the PPy–TiO₂–SnO₂ system, indicating oxide nanoparticle coalescence and improved structural ordering facilitated by the PPy matrix. Correspondingly, microstrain (ε) decreased substantially from 12.84 × 10^{−3} in TiO₂ to 4.20 × 10^{−3} in the tricomposite, while the dislocation density (δ) was reduced from 65.65 × 10¹⁴ m^{−2} to 7.72 × 10¹⁴ m^{−2}, signifying relaxation of lattice distortions and minimization of defect concentrations. Overall, the XRD results confirm that embedding TiO₂ and SnO₂ nanoparticles within PPy yields a more ordered, defect-reduced hybrid structure with larger crystallite domains and enhanced crystalline quality. Such structural refinement is expected to improve charge transport pathways, interfacial electron mobility, and surface reactivity making the PPy–TiO₂–SnO₂ tricomposite a promising candidate for advanced functional application of gas sensing.

The examination of UV-Visible Spectra revealed that doping resulted in an increase in the band gap of the material across all samples. The tricomposite material PPy–TiO₂–SnO₂ has a band gap of 4.26 eV, which is between the band gaps of 4.26 and 4.28 eV, makes it a unique gas sensing material. The impedance analysis confirmed non-Debye relaxation behaviour in all samples, evidenced by the absence of a well-defined semicircle and the presence of a linear Warburg diffusion region. From the Impedance analysis it has been concluded that equivalent R–(CR)–W circuit effectively captured the interfacial charge-transfer processes and ion-diffusion characteristics of the polymer–oxide systems. Among the composites, PPy–TiO₂ exhibited the highest Warburg impedance and charge-transfer resistance, indicating slower diffusion kinetics and reduced conductivity. This behaviour correlates with its relatively larger optical band gap, which limits the density of mobile charge carriers. Overall, the impedance findings reveal that the incorporation of metal oxides significantly modifies the electrochemical transport properties of PPy-based composites.

It has been analysed that the SEM micrographs of the PPy–TiO₂ composite exhibited a granular morphology characterized by well-connected spherical particles forming dense clusters, whereas the PPy–SnO₂ composite had significantly larger and more distinct spherical grains. The coexistence of conducting PPy with semiconducting SnO₂ and TiO₂ from EDX suggests the formation of active interfaces that are expected to promote interfacial charge transfer and improve sensor response.

The PPy–TiO₂–SnO₂ sensor represents a groundbreaking advancement in sensor technology, demonstrating functionality at remarkably low gas concentrations of up to 150 ppm, as evidenced by current research findings. The reduced recovery time of merely 76 s and a response time of 184 s position PPy–TiO₂–SnO₂ as an exemplary material for CO₂ gas sensing applications. We hypothesize that the electron accumulation layer may have facilitated the adsorption of carbon dioxide on the surface of the sensing materials. Conversely, the electron depletion layer elevates the potential energy barrier at the interface, resulting in an augmented initial resistance of the composites, hence improving the gas response. At a CO₂ concentration of 50 ppm, pure PPy exhibits a minimal sensitivity of 0.42, whereas the novel PPy–TiO₂–SnO₂ nanocomposite demonstrates a significantly higher intensity of 0.82.

The alteration of the optical energy band gap and its impact on sensor sensitivity indicates that the synthesized PPy-TiO₂ composite exhibits peak sensitivities of 0.91, 2.50, and 6.36 at CO₂ gas concentrations of 50, 100, and 150 ppm, respectively. This static response of the sensor attributed to the composite's electronic band gap of 4.28 eV, positioning it as the most sensitive option compared to other sensors. At the same time in order to have a best sensor the recovery time must be less hence dynamic response was obtained which stated that the novel PPy-TiO₂-SnO₂ has a response time of 184 s at 150 ppm which is just equivalent to the fast responding PPy-TiO₂ nanocomposite; but it takes very less recovery time of just 76 s in comparison with the recovery time of 112 s for PPy-TiO₂.

The PPy-TiO₂-SnO₂ sensor was evaluated for repeatability and reversibility at lower concentrations of 50 and 100 ppm CO₂ at room temperature concluded that it has an excellent intelligent sensing towards CO₂ without lagging time showcasing repeatability and reversibility. All samples were evaluated for thermogravimetric analysis (TGA), revealing that novel ternary composite PPy-TiO₂-SnO₂ exhibited significantly enhanced thermal stability, with negligible mass loss observed at elevated temperatures making it an excellent candidate for sensing applications.

In addition, the selectivity studies confirm that the PPy-TiO₂-SnO₂ sensor exhibits preferential response towards CO₂ compared to interfering gases such as NH₃, ethanol, and CO at 50 ppm. The sensor also demonstrates good long-term stability, with a sensitivity drift of only 6.10% over 35 days and minimal baseline variation. Furthermore, humidity-dependent measurements indicate stable sensing performance under varying RH conditions (30–70%), highlighting the robustness of the composite under practical environmental conditions. These results further validate the suitability of the developed sensor for reliable and real-world CO₂ sensing applications.

CRedit authorship contribution statement

Shrikrishna P. Yawale: Project administration, Supervision, Validation. **Deepika S. Yawale:** Methodology, Resources, Software. **Ashitosh P. Deshmukh:** Investigation, Methodology. **Aditya V. Tiwari:** Conceptualization, Data curation, Formal analysis, Funding acquisition, Writing – original draft. **Payal S. Dudhe:** Visualization, Writing – review & editing. **Shailesh P. Tiwari:** Resources, Software. **Shikhil S. Wanjari:** Software.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.nxnano.2026.100455](https://doi.org/10.1016/j.nxnano.2026.100455).

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